ZCAS UNIVERSITY

BACHELOR OF SCIENCE IN SECURITY AND CRIME SCIENCE

CIT2341 CRIME MAPPING

FINAL EXAMINATION 2025

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

DATE : 28 MAY, 2025.

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **NOT TURN THIS PAGE** until the invigilator tells them to do so.

SECTION A: COMPULSORY **(40 marks)**

QUESTION 1

Concept of crime mapping

Crime mapping is the process through which crime analysts and researchers use location information about crime events to detect spatial patterns in criminal activity. Crime information is the focus. A crime and disorder incident has a location where it occurred and a date and time when it occurred. Spatial thinking is defined as the knowledge, skills, and habits of mind to use concepts of space, tools of representation like maps and graphs, and processes of reasoning to organize and solve problems (Downs and de Souza 2005)

Crime mapping first began in the 1980's with the Chicago police department through spatial data analysis as a way to help them better understand the why, not just the where, behind crime. Before this there were simpler, less in-depth crime mapping measures taken that began this journey towards advanced crime mapping that we see today. Recently technology has become the primary focus for many police related developments. Concerns regarding the need for increased global security combined with budget cuts have further increased the push towards technology like Geographic information system (GIS).

Geographic information system advances have led to possibilities of change in the criminal justice field with there being a decrease in technology costs and increased available data, at the level of spaces and movement and the level of social issues such as political and demographic (Gemma, 2018). These advances have transformed mapping into a discipline in and of itself by establishing patterns; finding correlations to visualize the often-undetectable data (Galdon, Clavell & Frowd, 2014) that can be used at all levels of government, as well as at civil and private organizations (Gemma, 2018).

Studying the scenario above carefully, answer the following questions:

a) Describe the need for crime mapping. (4 marks)

b) It is said that "Crime is not randomly distributed". Discuss the meaning of this statement. (5 marks)

c) Describe the concept of crime hotspots. (4 marks)

d) There are certain factors that contribute to criminal behaviour, one of them is poverty. Discuss the effects of poverty on crime prevalence. (5 marks)

e) Discuss any five (5) reformatory activities that prisons and communities can set up to help youths, inmates and those released to reform. (10 marks)

f) Describe three (3) challenges of crime mapping. (6 marks)

g) Describe tree (3) activities tat may be done wen gathering evidence on a crime scene (6 marks)

[Total: 40 marks]

SECTION B: ANSWER ANY THREE (3) QUESTIONS **(60 marks)**

QUESTION 2

Crime has an inherent geographical quality. There are four (4) dimensions of crime as listed below, discuss them.

a) Legal (a law) (5 marks)

b) Victim (target) (5 marks)

c) Motivated offender (perpetrator) (5 marks)

d) Place (happens at a place) (5 marks)

[Total: 20 marks]

QUESTION 3

a) There are specific methods investigators use for collecting data about crime. Investigators usually employ more than one of these techniques to help ensure an accurate and comprehensive investigation process. Describe four (4) of these methods. (8 marks)

b) Today, criminal behavior in society is on the rise. An investigators' skills are the foundation of their career. With the rapid increase of security breaches, company hacks, and data leaks, cybercrime has become one of the most significant threats to global business. Skilled investigators are key for the safety of companies and governments institutions alike. Describe three (3) skills an investigator must possess. (6 marks)

c) Explain the importance of the following social institutions in deterring increase in criminal activities.

i) Family (3 marks)

ii) School (3 marks)

[Total: 20 marks]

QUESTION 4

a) There are two (2) common types of data analysis methods. With the use of two (2) examples in each case, discuss these methods:

i) Quantitative method (5 marks)

ii) Qualitative method (5 marks)

b) The following is a description of the major components of a GIS, describe them

i) Data representation, (2 marks)

ii) Data features, (2 marks)

iii) Visualization, (2 marks)

iv) Scale, and (2 marks)

v) Querying. (2 marks)

[Total: 20 marks]

QUESTION 5

a) Problem solving is a methodical process for reducing the impact of crime and disorder problems in a community. The problem-solving approach is an integral component of the philosophy of community policing. The **SARA** approach used in problem solving is described as:

i) Scanning (S), (4 marks)

ii) Analysis (A), (4 marks)

iii) Response (R), and (4 marks)

iv) Assessment (A). (4 marks)

Using one (1) example in eac case, describe these stages.

b) Describe two (2) data integrity issues associated with tabular and geographic law enforcement data. (4 marks)

[Total: 20 marks]

END!!!